

VLAD THE STAKER

~ or ~

how to stop burning



A long time ago on a platform far,
far away....

RICHARD CRAIB

*made a comment that stuck with me
people don't spend enough time
thinking about stake allocation*

STAKE ALLOCATION == STOCK ALLOCATION

If you think of the tournament as the stock market...

1. You have a plethora of models to choose from
2. Your problem is allocating the right amount of NMR
3. You can long or short (P , $1-P$) each model

but it feels bad to short your own models if you trained them to be good!

4. Performance measured against $0.5 \times \text{CORR} + 2 \times \text{MMC}$
5. MMC is a moving target

STAKE ALLOCATION == STOCK ALLOCATION

Development history of Vlad

1. Stake on 4-5 models based on what I thought was a good idea (**bad idea**)
2. Pick 4-5 models that I like, and find an optimal NMR allocation based on model history by brute forcing different stake allocations (**uninformed**)
3. Use the R package fPortfolio and generate “robust” time series based on covariance matrices (**complex, breaks**)
4. Use the R package fPortfolio on a timeseries resampled from your model histories (**simple, works**)

So, in more detail,
what is Vlad doing?



VLAD THE STAKER – DATA INPUT

- Vlad reads in an excel sheet with the models you want to have included in your stake optimization, and the starting era of the model.
- Here I start at a minimum of 339, as that is also the start of the daily tournament. Any models that are <60 rounds old are disregarded.

	A	B
1	ModelName	Starting Era
2	NB_EXAMPLE_MODEL	602
3	NB_FEAT_NEUTRAL	602
4	NB_HELLO_NUMERA1	553
5	NB_TARGET_ENSEMBLE	602
6	V2_EXAMPLE_PREDS	339
7	V3_EXAMPLE_PREDS	339
8	V4_EXAMPLE_PREDS	339
9	V4_LGBM_JEROME20	339
10	V4_LGBM_JEROME60	339
11	V4_LGBM_NOMI20	339
12	V4_LGBM_NOMI60	339
13	V4_LGBM_RALPH20	339
14	V4_LGBM_RALPH60	339
15	V4_LGBM_TYLER20	339
16	V4_LGBM_TYLER60	339
17	V4_LGBM_VICTOR20	339

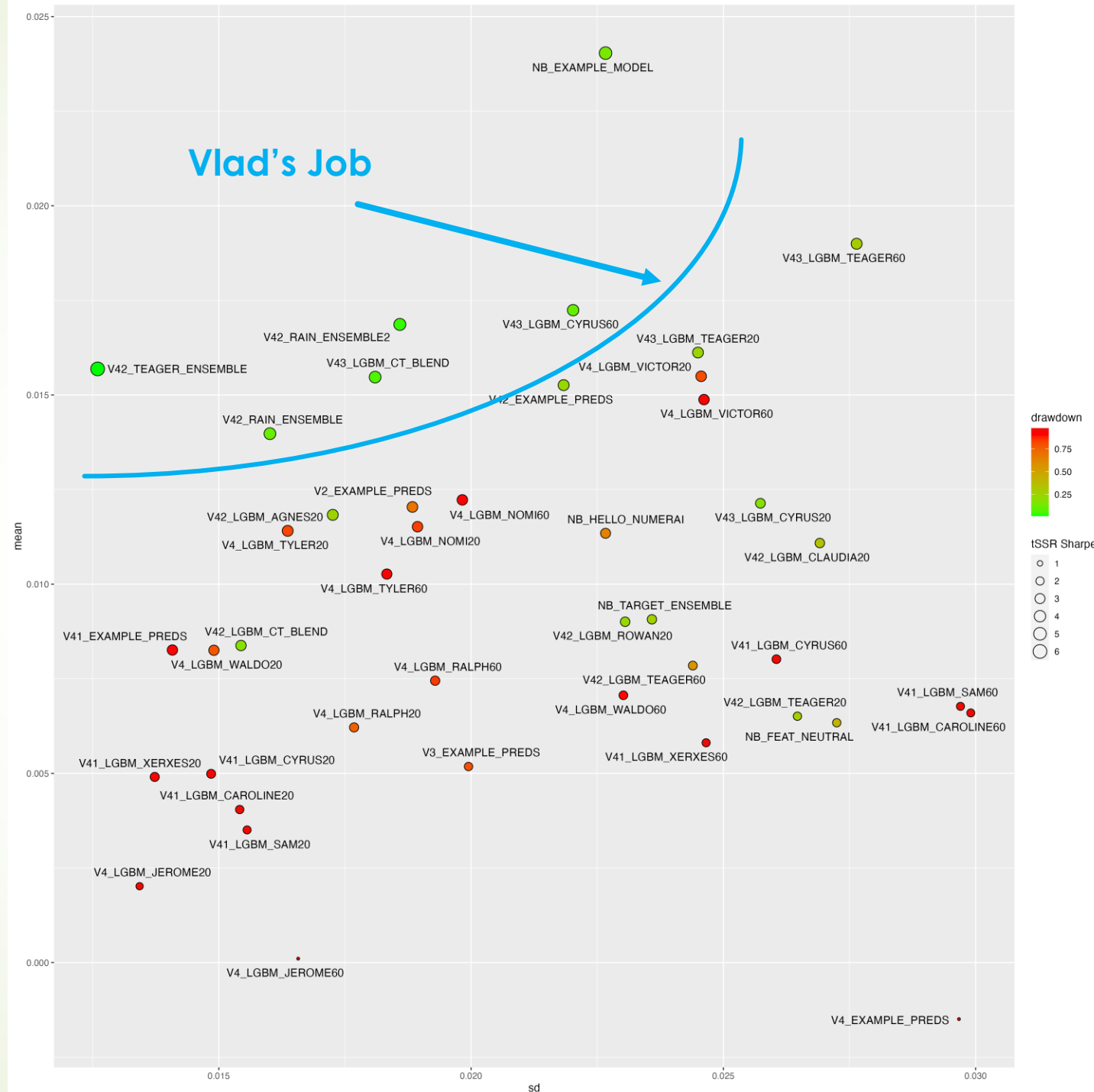
MODEL PERF.

- Vlad calculates a number of model stats to give you a graphical overview of your models.
- We have the **mean score** on the y-axis, **standard deviation** of the score on the x-axis.
- Color is the **max drawdown** (red == bad)
- Size is tSSR: a measure of sharpe that is corrected for sample size and autocorrelation (bigger == better)



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STEP 1. SUBSETTING

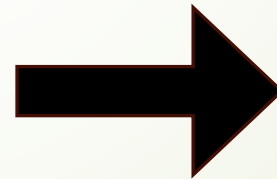
- Vlad only considers models with a mean > 0.0
(the fPortfolio algorithms sometimes have trouble with negative-scoring assets)
- Vlad filters the dataset for days where we have information on all models

STEP 2. RESAMPLING

I want stability. I don't want to drastically redistribute my stakes every week

Although with dedicated stakeslots, stake management hurts less

- Longer time series work better
- But resampling the time series you have 40 times works as well

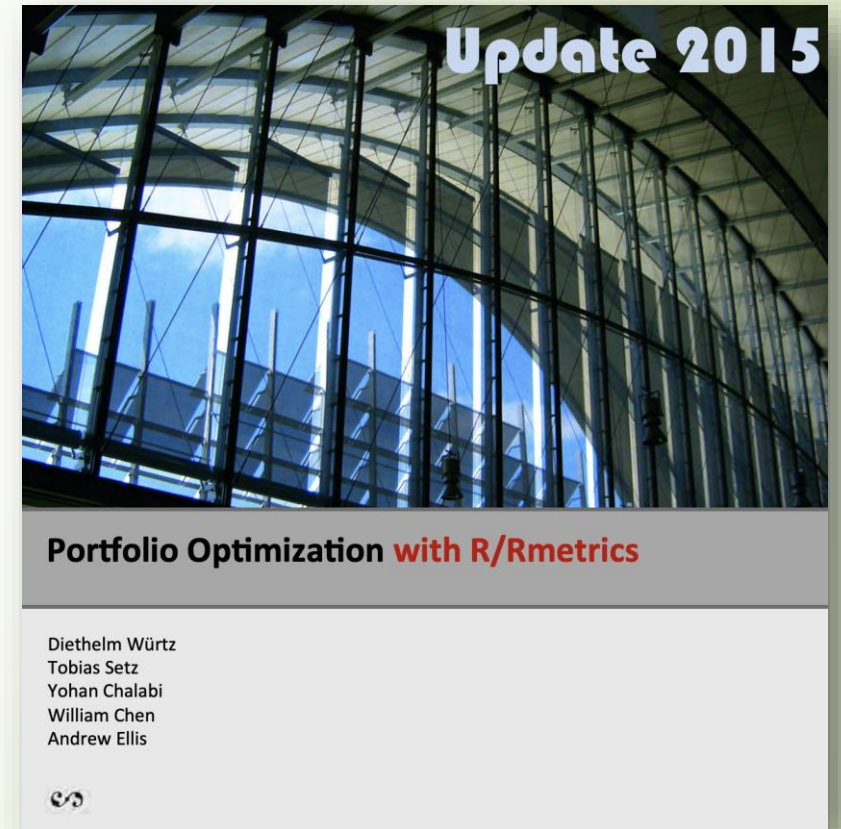


Vlad resamples

STEP 3. PICK A PORTFOLIO STRATEGY

There are a bunch of different goals you can specify when optimizing portfolios.

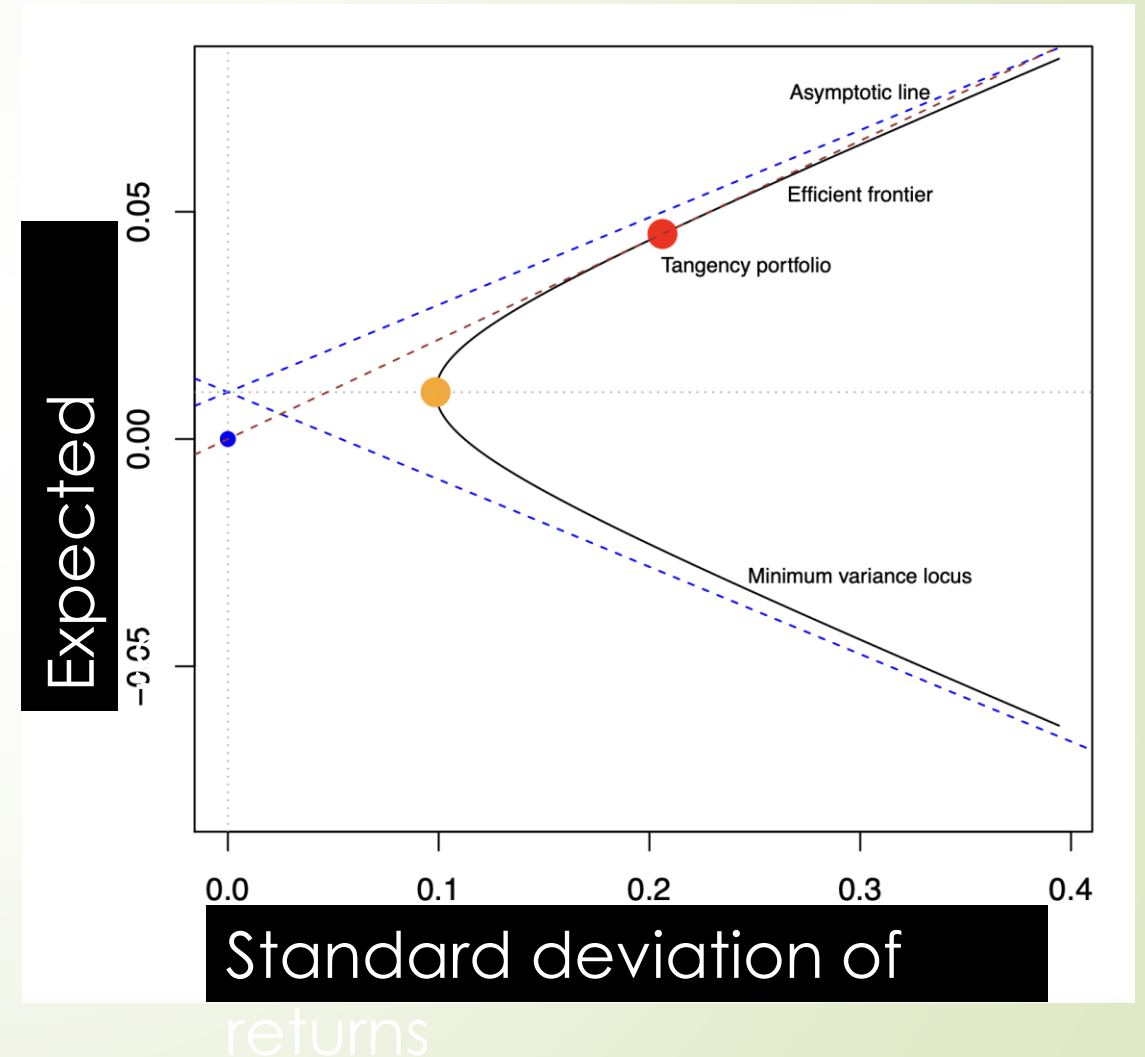
(and this is not my field at all)



STEP 3. PICK A PORTFOLIO STRATEGY

Vlad averages between the red dot (tangency portfolio), and the orange dot (minvariance portfolio)

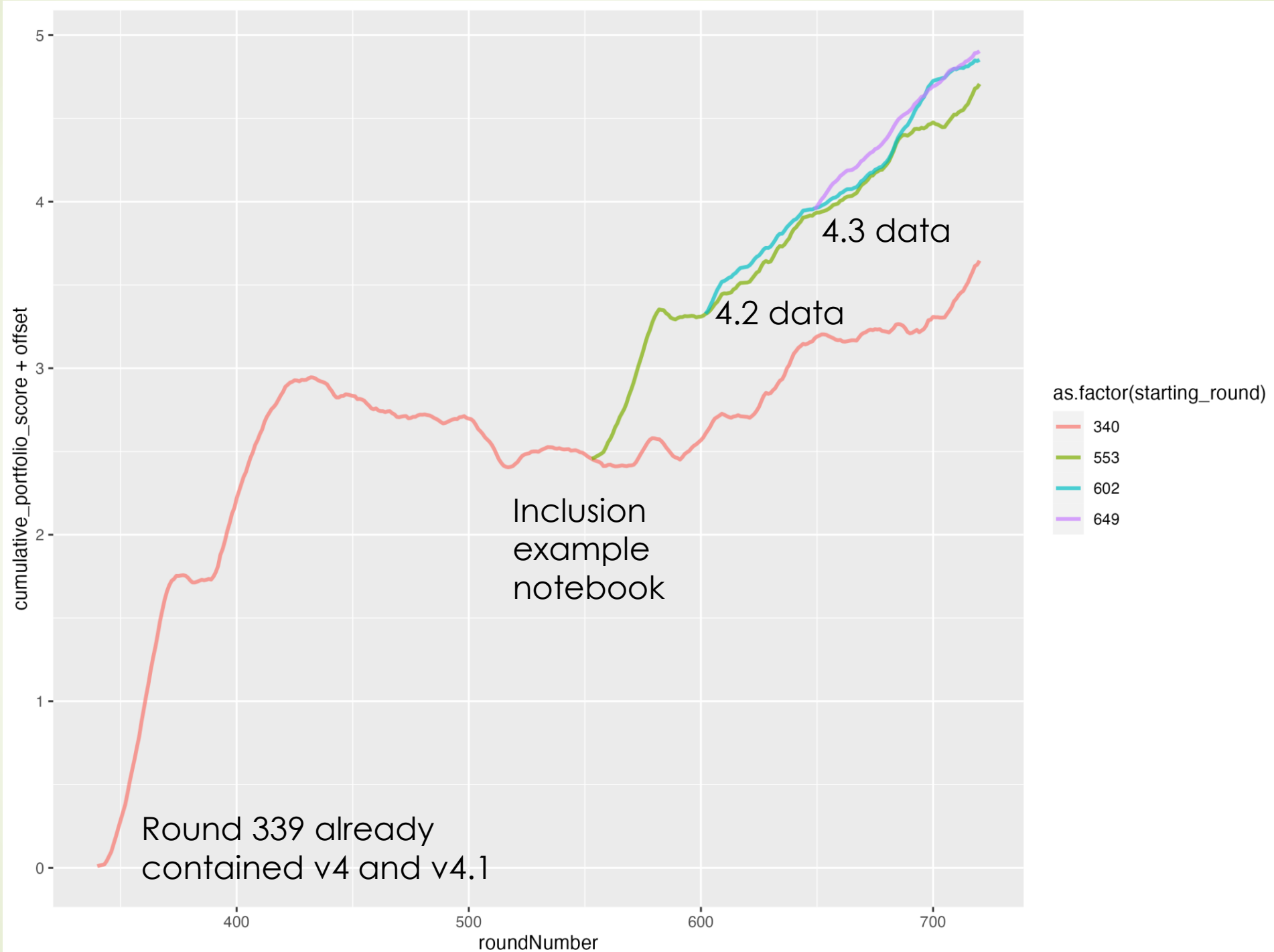
Both are points on the efficient frontier - a theoretical description of the tradeoff between higher returns and higher variance



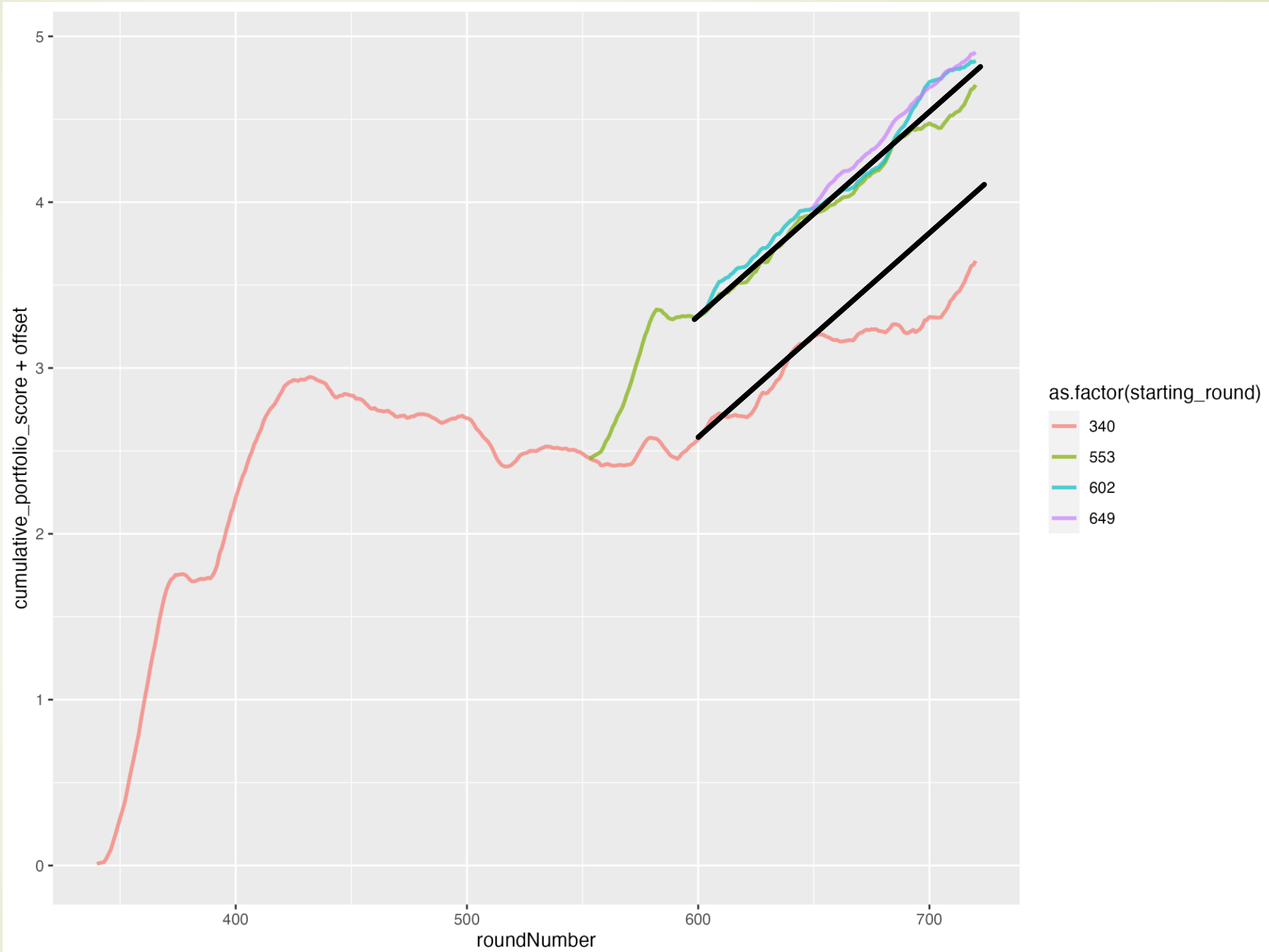
VLAD OUTPUT - TABLE

lname	lweight	lstake	lmean	lCov	lCVaR	lVaR	lsamplesize	lstarting_round
IV2_EXAMPLE_PREDS	0.227	568	0.0096	0.0215	0.0213	0.0169	378	340
IV3_EXAMPLE_PREDS	0.257	642						
IV41_LGBM_SAM20	0.196	490						
IV4_LGBM_VICTOR20	0.319	798						
INB_HELLO_NUMERAI	0.099	248	0.0136	0.0138	0.0104	0.0063	165	553
IV41_LGBM_CYRUS20	0.198	495						
IV41_LGBM_XERXES20	0.214	535						
IV4_LGBM_NOMI60	0.152	380						
IV4_LGBM_RALPH60	0.095	238						
IV4_LGBM_VICTOR60	0.132	330						
IV4_LGBM_WALDO60	0.109	272						
INB_EXAMPLE_MODEL	0.155	388	0.0133	0.0091	0.0017	-0.0003	116	602
IV41_LGBM_CAROLINE20	0.127	318						
IV41_LGBM_XERXES20	0.147	368						
IV42_EXAMPLE_PREDS	0.135	338						
IV42_RAIN_ENSEMBLE	0.147	368						
IV42_RAIN_ENSEMBLE2	0.146	365						
IV4_LGBM_VICTOR60	0.143	357						
INB_EXAMPLE_MODEL	0.195	488	0.0137	0.006	-0.0023	-0.0042	71	649
INB_TARGET_ENSEMBLE	0.112	280						
IV41_EXAMPLE_PREDS	0.125	312						
IV42_LGBM_TEAGER60	0.118	295						
IV42_RAIN_ENSEMBLE	0.107	268						
IV43_LGBM_CYRUS60	0.126	315						
IV4_LGBM_NOMI60	0.119	298						
IV4_LGBM_TYLER20	0.099	248						

VLAD OUTPUT - TABLE



VLAD OUTPUT - TABLE



VLAD IS WAITING FOR YOU

github:
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/vladthetaker

